Hyperplane Arrangements and Combinatorial Complexity

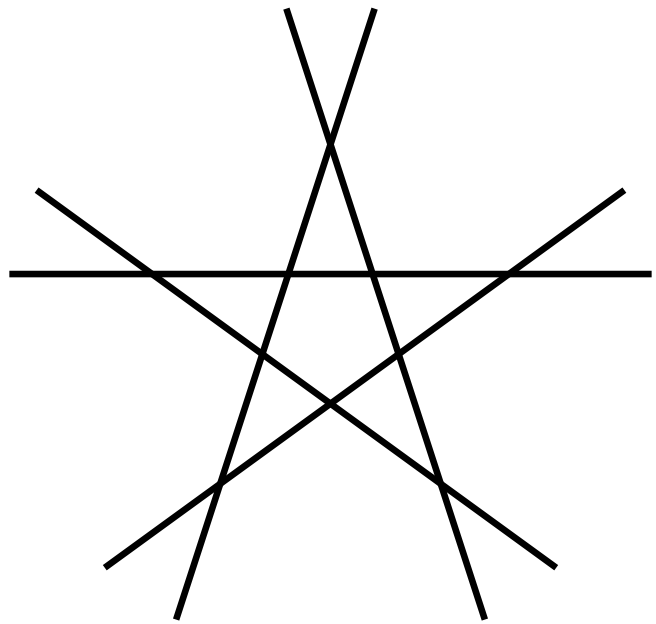
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What is a Hyperplane Arrangement?

A *hyperplane arrangement* $\mathcal{A}$ in $\mathbb{R}^d$ is a finite set of affine hyperplanes:

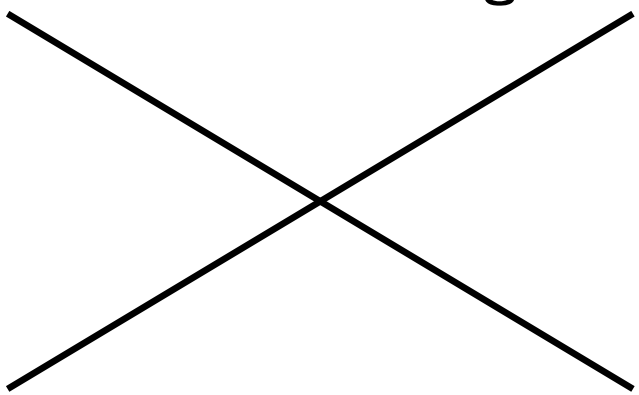
$$\mathcal{A} = \{H_1, \ldots, H_n\}, \qquad H_i = \{x \in \mathbb{R}^d : w_i \cdot x + b_i = 0\}.$$



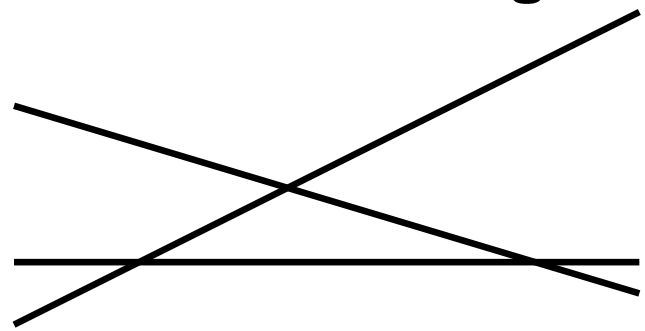
- Even a few hyperplanes create interesting region structure.
- These regions encode combinatorial data.

Examples of Arrangements

Two lines = 4 regions



Three lines = 7 regions



Sign Patterns

For hyperplanes $H_i : w_i \cdot x + b_i = 0$, define

$$\sigma(x) = \big(\text{sign}(w_1 \cdot x + b_1), \ldots, \text{sign}(w_n \cdot x + b_n)\big).$$

Key fact

Two points lie in the same region $\iff$ they have the same sign pattern.

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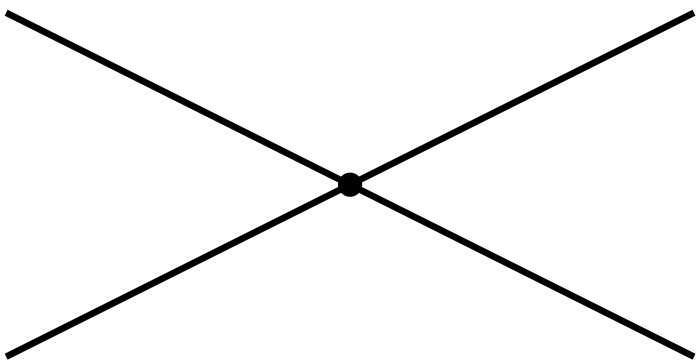
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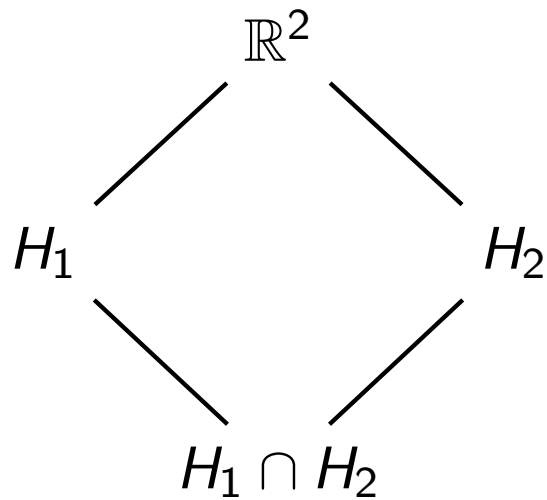
- Regions correspond to consistent choices of $+$ and $-$ for each hyperplane.

Intersection Poset

The *intersection poset* $L(\mathcal{A})$ contains all intersections of hyperplanes, ordered by reverse inclusion.

Example: Two lines





Characteristic Polynomial

Defined by

$$\chi_{\mathcal{A}}(t) = \sum_{X \in L(\mathcal{A})} \mu(\mathbb{R}^d, X)\, t^{\dim X}.$$

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$$\chi(-1) = 4 = \text{number of regions.}$$

Zaslavsky's Theorem

$$r(\mathcal{A}) = (-1)^{\mathsf{rank}(\mathcal{A})}\,\chi_{\mathcal{A}}(-1)$$

$$b(\mathcal{A}) = (-1)^{\mathsf{rank}(\mathcal{A})}\,\chi_{\mathcal{A}}(1)$$

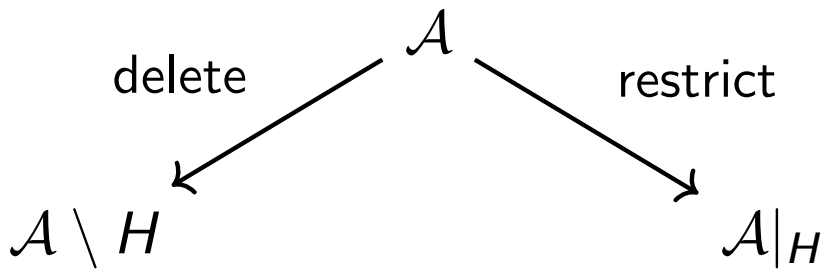
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Idea of proof

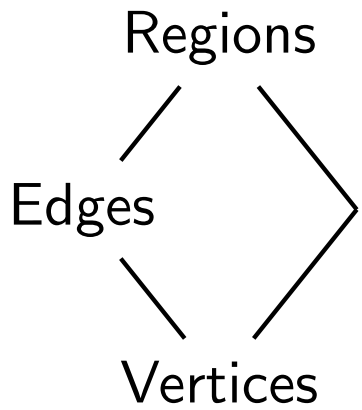
Deleting a hyperplane or restricting to it gives a recurrence for region counts. The characteristic polynomial satisfies the same recurrence. Conclude by induction.

Deletion–Restriction Diagram



$$r(\mathcal{A}) = r(\mathcal{A} \setminus H) + r(\mathcal{A}|_H)$$

Face Poset



- Faces are ordered by *closure inclusion*: $F \leq G \iff \overline{F} \subseteq \overline{G}$.
- Maximal elements are the regions.
- Encodes adjacency and how the stratification fits together.

Sign Vectors and Adjacency

$$\sigma(x) \in \{-, 0, +\}^n$$

- Zero entries: point lies on a hyperplane.
- No zeros: interior of a region.
- Adjacent regions differ in exactly one sign coordinate.

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This is the oriented structure of the arrangement.

ReLU = Hyperplane Arrangement

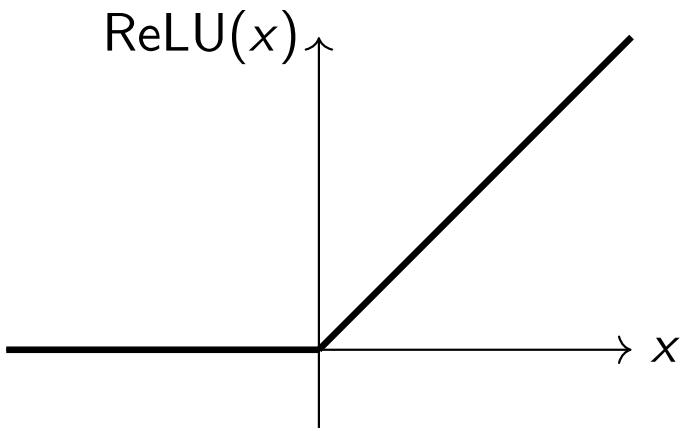
A ReLU unit computes:

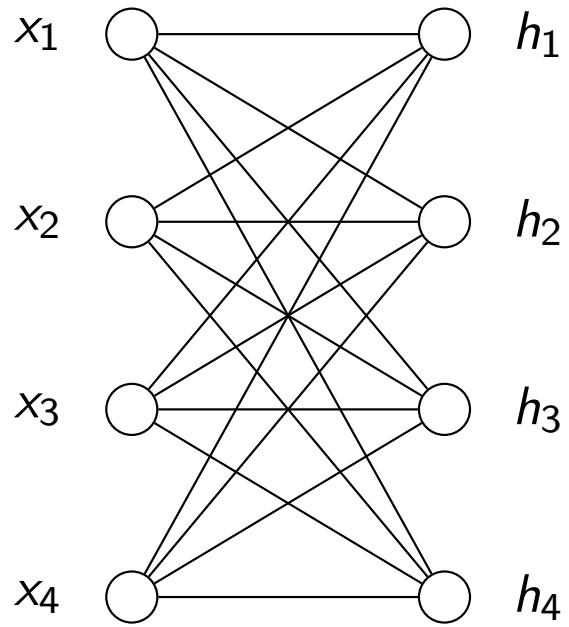
$$\mathrm{ReLU}(w \cdot x + b) = \max(0,\ w \cdot x + b).$$

Its activation boundary is the hyperplane

$$w \cdot x + b = 0.$$

A layer with n ReLU units in $\mathbb{R}^d$ produces n such hyperplanes.





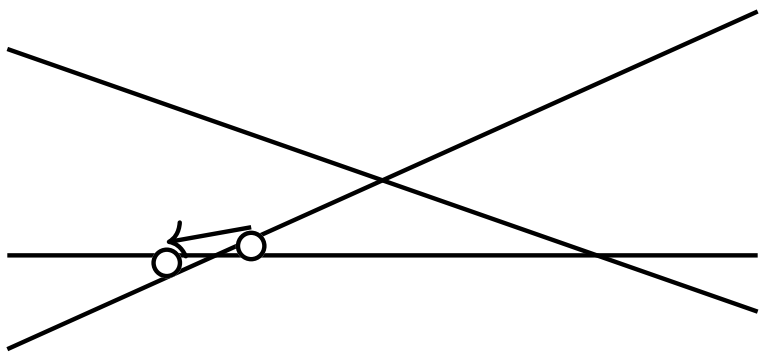
Linear Regions of a ReLU Layer

- On each region, all signs are fixed.
- Therefore the layer is *linear* on each region.
- Regions = linear pieces.

$$\#\text{linear pieces} = r(\mathcal{A}) = (-1)^d \chi_{\mathcal{A}}(-1).$$

Fragility and Decision Boundaries

- Near intersections of several hyperplanes, even tiny perturbations can move an input into a different region.
- Crossing multiple boundaries flips multiple ReLUs at once, changing the affine function applied.
- This offers a geometric explanation for adversarial sensitivity.



Conclusion

- Hyperplane arrangements provide a clean combinatorial description of region structure.
- Zaslavsky's theorem turns region counting into a poset computation.
- ReLU networks naturally inherit this structure: regions correspond to linear behaviors of the layer.

References



A. Chen and A. Wang, *Intersection Posets, Characteristic Polynomials, and Regions*, MIT PRIMES Conference (2017). https://math.mit.edu/research/highschool/primes/materials/2017/conf/Chen-Wang.pdf